

Design and Development of Mobile Healthcare Application Prototype Using Flutter

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Abstract— with the increased awareness of people to the importance of regular health checkup, healthcare-providing facilities such as hospitals and clinics are struggling to cope with the demands of an ever-increasing population. This problem is even worse in countries with weak healthcare facilities. The overload on hospitals affects the quality of healthcare services. Many patients spend long hours searching for prescribed medications; others wait for long times to get appointment to see their doctors. Access to ambulance services in critical cases can also be very difficult. All of these factors in addition to the high cost of treatment at hospitals make remote delivery of healthcare the most practical option for many patients. In this paper, a healthcare mobile application prototype was developed successfully and installed on an Android smartphone. The application offers convenient telemedicine services that enable patients to be diagnosed online by medical specialists. The application also facilitates the ordering of medications using online payment. Medications can be delivered directly to the desired location, and in emergency cases, patients can call for ambulance service. The system enables medical doctors and nurses to monitor their patients' health continuously. Pharmacies can also use the application to offer and deliver their medical products to patients.

Keywords—Telemedicine, Healthcare Application, Flutter, Demo Application, Ecosystem

I. INTRODUCTION

Telemedicine is a medical and public health practice which is supported by mobile devices, wireless telecommunication systems, and impeded patient monitoring devices [1]. Thanks to the recent advancements in smartphones in terms of speed and storage coupled with wide coverage and availability of Wi-Fi internet, various types of smartphone healthcare application have been developed [2].

The increased public awareness of disease prevention through healthy life style, has led to the development of many mobile health applications that are widely available in online public application stores. The two dominant mobile systems platforms, Android and iOS, host more than 165,000 mobile applications for health monitoring, out of which 9% provide screening, diagnosing and monitoring of a broad spectrum of illnesses[3]. Examples of these applications are: MDLIVE, Lemonaid, Doctor on Demand, Amwell, Talkspace, Teladoc, BCBSM Online Visits, Spruce, DocsApp[4].

Mobile health devices and applications provide many benefits for Healthcare professionals by supporting better

clinical decision-making and improving patient outcomes. However, some healthcare professionals remain reluctant to adopt their use. Despite the benefits offered by mobile health apps, better standardization and validation practices need to be established to ensure the proper use and integration of these increasingly sophisticated tools into medical practice. [6-8].

The recent Covid-19 pandemic has made mobile health applications very popular. Fortunately, there are many programming tools that are available to application developers. Flutter is one of these tools, which has distinguished itself lately. Flutter is an open-source UI software development kit created by Google[5].

In this paper we have used Flutter to develop an integrated mobile healthcare application that enables reliable, fast, and most importantly affordable healthcare delivery application.

II. METHODOLOGY

A. THE CONCEPT OF THE MOBILE HEALTHCARE APPLICATION

The following diagram shows the main components of the proposed healthcare delivery application.

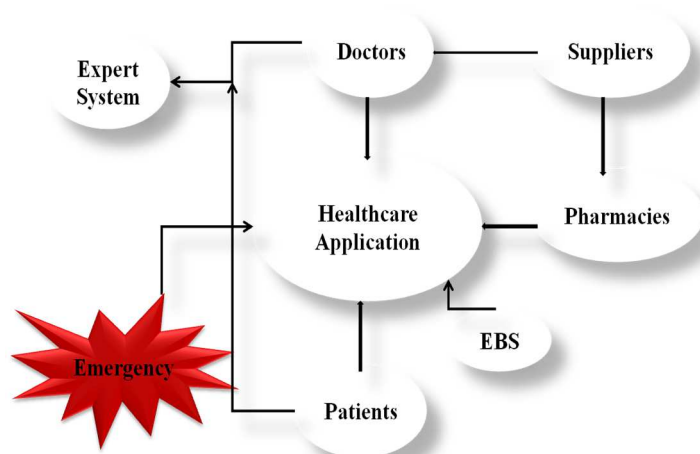


Figure 1: The Concept of the Proposed Mobile Healthcare Application

As shown in Figure 1, the application provides patients an easy access to doctors and healthcare providers anywhere in the world through immediate consultancy by text message,

images and scans or even real-time video calls. Prescribed medications can be ordered from multiple pharmacies connected to the application, and delivered to the desired location after completing the payment through electronic banking services (EBS). The system includes a booking system that enables patients to book for online sessions with their doctors to discuss their symptoms. In emergency cases, the application will allow the person accompanying the patient to call for an ambulance while receiving advice from the doctor on how to stabilize the patient condition.

The application also provides a platform for medical devices/medications suppliers to make their products available directly to doctors and pharmacists. Furthermore, data related to the cases can be stored in an expert system for education and training purposes.

Ambulance ordering through the application portal will ease collection of patient's initial information in the cases of chronic diseases and allergies, which can be sent directly to the hospital information system. Emergency doctors can also follow-up with the co-patient to help stabilize the patient's condition.

B. PROTOTYPE DEVELOPMENT

The application prototype was developed according to the following phases:

Phase 1: Selecting the System components



Figure 2: Healthcare application Components

As shown in Figure 2, the prototype was developed using a combination of application tools, which include Flutter, Android Studio and Plugins.

Flutter is a new cross platform mobile application SDK offered by Google which assists developers to build various Android and iPhone Operating System (iOS) applications using a common codebase. Apps built using Flutter are written in Dart language. Flutter uses modern reactive framework allowing developers to build awesome UI with native look and feel. Google says, it took React as Inspiration to develop Flutter. Flutter under the hood uses a 2D rendering engine to build all the UI elements. Thus, apps built in Flutter look the same on Android and iOS [9]. The framework of Flutter consists of Dart, as the underlying programming language, Flutter Engine as the rendering support layer, Foundation Library for providing basic classes and functions for the application, Widgets and visual blocks including design-

specific element sets like Google's Material Design and Human/System Interface Guidelines.

Android Studio is the official integrated development environment (IDE) for Android application development. It is based on the IntelliJ IDEA, a Java integrated development environment for software, and incorporates its code editing and developer tools. It supports application development within the Android operating system, Android Studio uses a Gradle-based build system, emulator, code templates, and GitHub integration. Every project in Android Studio has one or more modalities with source code and resource files. These modalities include Android application modules, library modules, and Google Application Engine modules.

Flutter's Plugins & Dart's Plugins

Flutter plugin is the wrapper of the native code like android (Kotlin or java) and iPhone Operating System (iOS) (swift or objective C). It can do anything that a native application by using Platform Channels and Message Passing. Also, it instructs the native iOS/Android code to perform actions and returns the result to Dart [10].

The Dart plugin adds Dart support to JetBrains IDEs such as IntelliJ IDEA and Android Studio. IntelliJ IDEA is an intelligent Java IDE with support for many other languages and frameworks. Android Studio is an IDE based on IntelliJ IDEA that's used for Android and Flutter development [11].

Phase 2: System Installation and Setup

The Flutter SDK, and Android Studio were downloaded, linked together, then Flutter's and Dart's plugins were downloaded as well, and the main Dart was executed after enabling the Android Emulator.

Phase 3: Application System Architecture

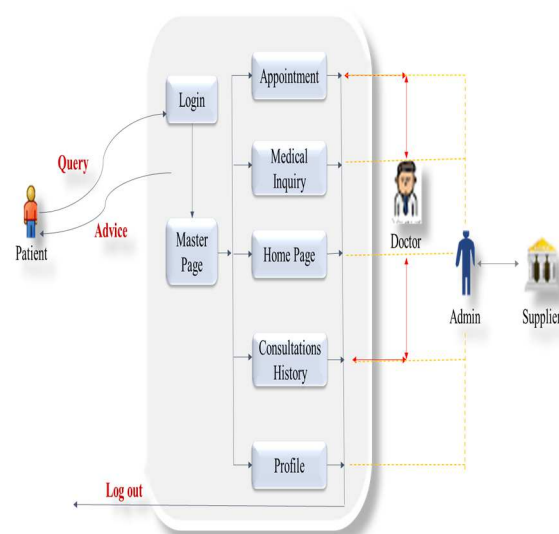


Figure 3: UML Use Case Diagram

In Figure 3, the unified modeling language (UML) Use Case Diagram for the prototype is shown, where the patient

can login into the application homepage with a username and a password, which is passed to the master page validation and then redirected to the rest of the pages. In the profile page, patients can upload their personal information such as age, gender, address and contact phone number. While, in the Home page, they can interpret provide their symptoms to all doctors or to a specific specialist. The doctor can reply through the consultation history page, and in case of medications prescription, they can order them using the medical enquiry page by deciding whether they want it to be delivered once or monthly (mainly for chronic diseases such as diabetes and hypertension) and give the desired delivery location. In the appointments page, patients can see a list of doctors (categorized based on their specialty) to review their profile and patient's remarks, and book an appointment.

Phase 4: Development and Testing of the Proposed Application

To lunch the healthcare application, the Android Studio was used to start a new Flutter project by clicking the start button. The project was named SHIFAA, and company domain was inserted, to identify the application for future website implementation, then a sample application (scaffold) was generated to see how the application might look like. This is followed by the pressing the finish button. The starting code to create a simple counter application was executed, using Android Emulator. Then a material application was developed and linked with the widget home to insert the programming code.

Phase 5: Installation on Mobiles and Final Testing

The Android Application Package (APK) file extension in the Open Handset Alliance, which is a mobile phone platform, based on Linux, created when using Google's Android software development kit (SDK). It contains the code files, resource files and the Android Manifest XML file extension. [12].To deploy the APK on an Android smartphone, the developer mode in the phone was first enabled followed by enabling USB debugging in the development settings to allow communication with the computer. Alternatively, the build number can be tapped seven times then the phone is linked to the computer using USB, and finally the application can be accessed from android studio.

III. RESULTS AND DISCUSSION

Screenshots of the proposed healthcare application are given in the following sections to show the perspectives from patient's point of view.

1) Uploading the APK on the Android Platform

Figure 4, shows the prototype demo APK on an Android smart phone.

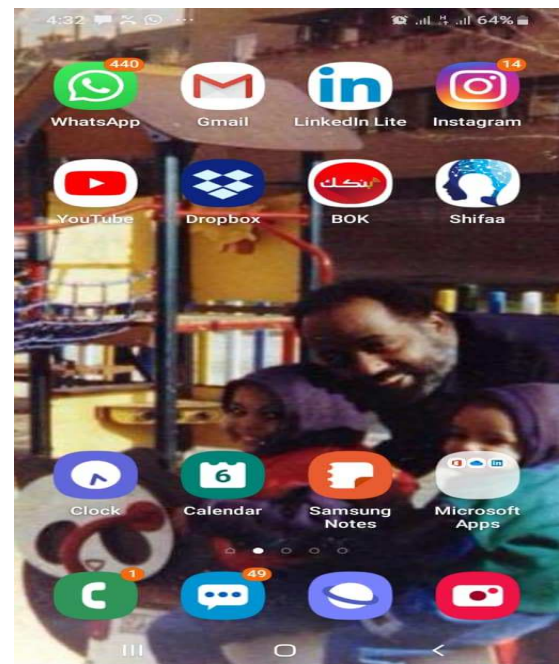


Figure 4: Demo APK in Android Mobile

2) Logging into the application

Figure 5 shows the login screen with a username and password.

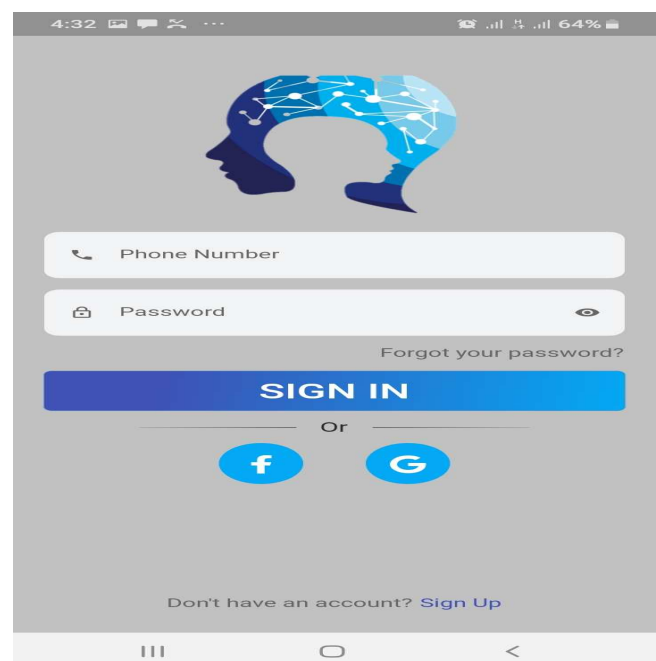


Figure 5: Logging into the Application

3) Application Home Page

Figure 6 shows where patients describe their symptoms to all available doctors or to one specific doctor.

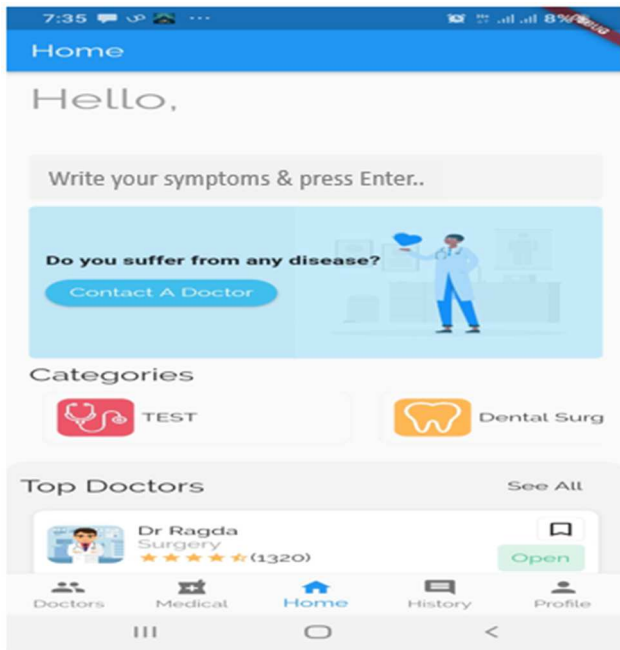


Figure 6: Home Page

4) Patient Data Entry

Figure 7 shows the interface for patients to save their personal information, to make it easy for doctors to contact them and for the delivery company know where to deliver medications.

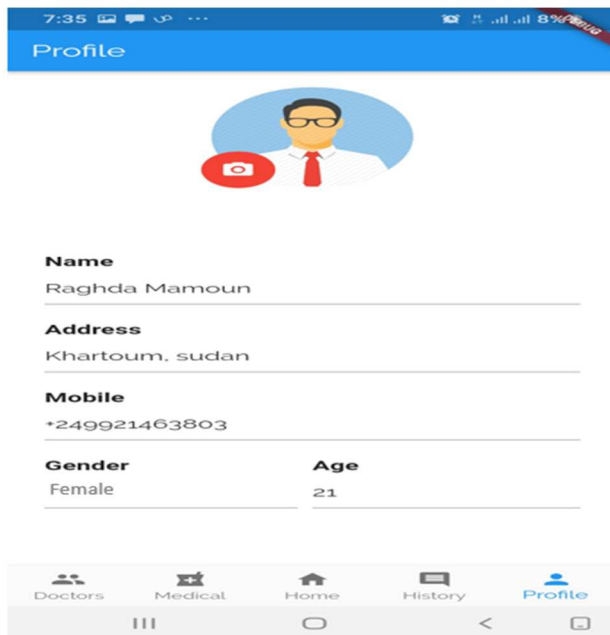


Figure 7: Patient's Profile

5) Consultations History

This is where all the consultations related to the patient will be kept.

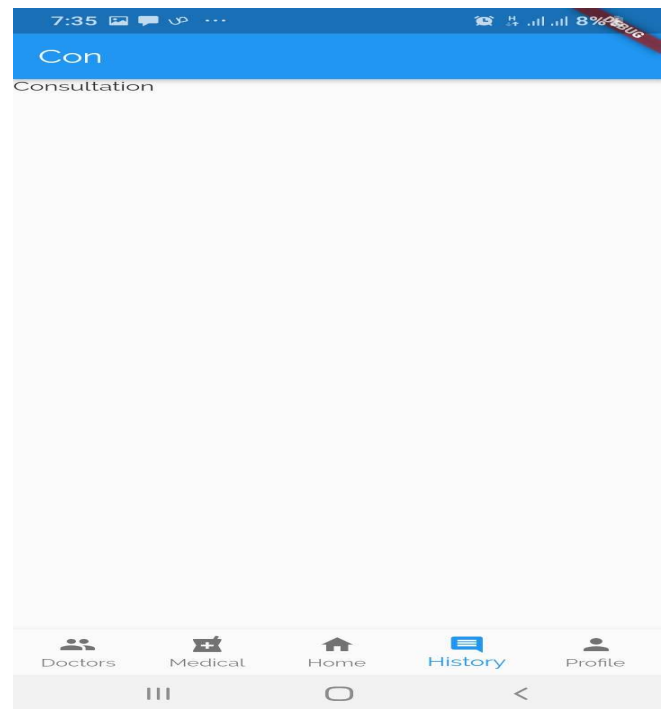


Figure 8: Consultations History

6) Precribed Medication Ordering

Figure 9 shows the window where patients can order medications, once or monthly, pay online, and have them delivered to the patient's desired location.

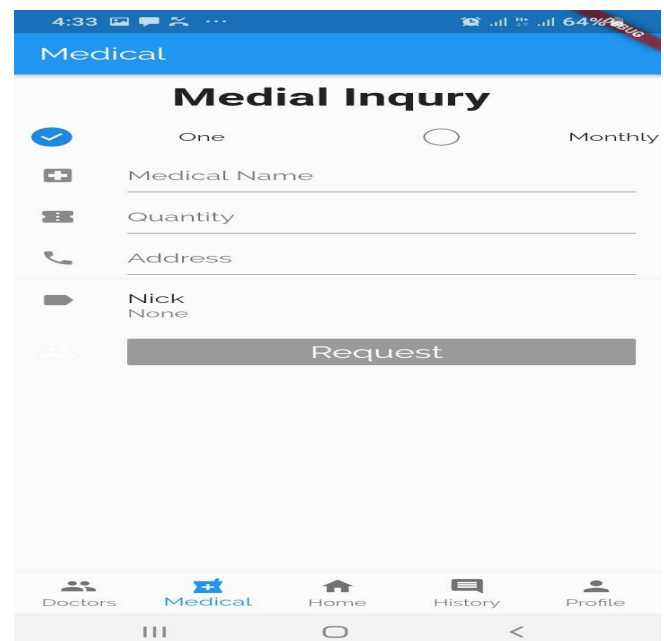


Figure 9: Medications Ordering Window

7) Avaible Doctors Information

Figure 10 shows the online page for the list of doctors available for patients for making booking and appointment.

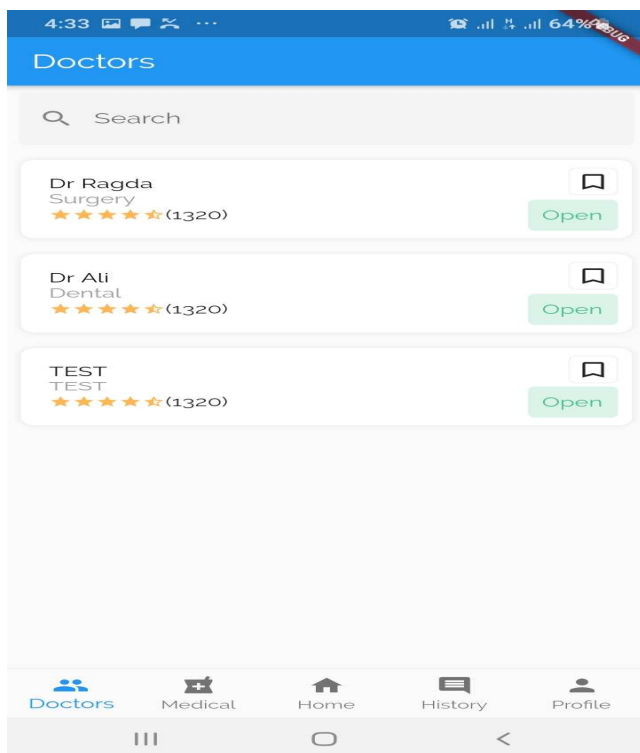


Figure 10: Doctors List

8) Appointment Booking

Patients can choose which doctor to consult or book an appointment with based on their specialties and expertise stated in their profiles.

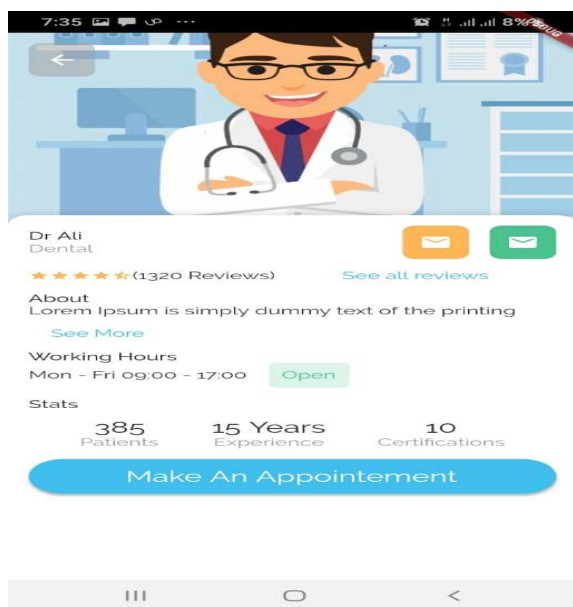


Figure 11: Doctor Profile /Appointment Booking

IV. CONCLUSION

The quality of healthcare services delivery is affected by the overload in hospitals in Sudan, resulting in an actual need for telemedicine services. A healthcare delivery mobile application prototype was developed successfully using Flutter, and launched on an Android smart phone. The application can also be deployed on other types of smart phones to offer convenient telemedicine services that

allow patients to get diagnosed online by doctors from anywhere in the world, order medications and make online payment. Also, in emergency cases, patients can call for ambulance services. The patient data privacy scheme was not included in the prototype, but definitely it will be dealt with when developing the final application. The performance of the application will be evaluated after being used by a fairly large number of patients, healthcare providers and medication suppliers.

V. FURTHER ENHANCEMENT TO BE INCLUDED IN THE FINAL VERSION OF THE APPLICATION

The proposed mobile healthcare application can be enhanced further by:

1. Including doctors' portal and administrative control console in final version of the application.
2. Configuring the application to work on both Android and iOS smartphones.
3. Including alternative solutions in case there was no Internet access.
4. Including artificial intelligence impeded systems, augmented reality (AR) features to make the experience more realistic, and an expert system that will keep doctor's precious expertise and clinical cases in an expert System, to help in training the AI network in the future for clinical decision making.
5. Including an interface that connects suppliers with pharmacies and doctors.
6. Conducting performance analysis and evaluation sometime after launching the application.

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